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1. A function's single most important role is to

   a. give a name to a block of code.

   b. reduce program size.

   c. accept arguments and provide a return value.

   d. help organize a program into conceptual units.

Ans: d. help organize a program into conceptual units.

2. A function itself is called the function d**<u>Definition</u>**.

3. Write a function called foo() that displays the word foo.

Ans:

```
#include<iostream>
using namespace std;

void foo(){
     cout << "foo";
}
```

4. A one-statement description of a function is referred to as a function **<u>Declaration</u>** or a **<u>Prototype</u>**.

5. The statements that carry out the work of the function constitute the function **<u>Body</u>**.

6. A program statement that invokes a function is a function **<u>Call</u>**.

7. The first line of a function definition is referred to as the **<u>Heading (Function Header)</u>**.

8. A function argument is

a. a variable in the function that receives a value from the calling program.

b. a way that functions resist accepting the calling program's values.

c. a value sent to the function by the calling program.

d. a value returned by the function to the calling program.

Ans: c. a value sent to the function by the calling program.

9. True or false: When arguments are passed by value, the function works with the original arguments in the calling program.

Ans: False

10. What is the purpose of using argument names in a function declaration?

Ans: To make the function declaration more readable and indicate the purpose of each parameter. The names are optional.

11. Which of the following can legitimately be passed to a function?

a. A constant

b. A variable

c. A structure

d. A header file

Ans: a. A constant b. A variable c. A structure

12. What is the significance of empty parentheses in a function declaration?

Ans: It means the function takes no arguments.

13. How many values can be returned from a function?

Ans: One value

14. True or false: When a function returns a value, the entire function call can appear on the right side of the equal sign and be assigned to another variable.

Ans: True

15. Where is a function's return type specified?

Ans: Before the function name in the function declaration or definition.

16. A function that doesn't return anything has return type **void**.

17. Here's a function: int times2(int a) { return (a*2); } Write a main() program that includes everything necessary to call this function.

Ans:
```
#include<iostream>
using namespace std;
int times2(int a){
     return (a*2);
}
int main(){
     int result;
     result = times2(5);
     cout << "Result = " << result;
     return 0;
}
```

18. When an argument is passed by reference

a. a variable is created in the function to hold the argument's value.

b. the function cannot access the argument's value.

c. a temporary variable is created in the calling program to hold the argument's value.

d. the function accesses the argument's original value in the calling program.

Ans: d. the function accesses the argument's original value in the calling program.

19. What is a principal reason for passing arguments by reference?

Ans: To allow the function to modify the original variable without making a copy.

20. Overloaded functions

a. are a group of functions with the same name.

b. all have the same number and types of arguments.

c. make life simpler for programmers.

d. may fail unexpectedly due to stress.

Ans: a. are a group of function with the same name.

c. make life simpler for programmers.

21. Write declarations for two overloaded functions named bar(). They both return type int. The first takes one argument of type char, and the second takes two arguments of type char. If this is impossible, say why.

Ans:
```
int bar(char);
int bar(char, char);
```

22. In general, an inline function executes **Faster** than a normal function, but requires **more** memory.

23. Write the declarator for an inline function named foobar() that takes one argument of type float and returns type float.

Ans: `inline float foobar(float);`

24. A default argument has a value that

a. may be supplied by the calling program.

b. may be supplied by the function.

c. must have a constant value.

d. must have a variable value.

Ans: b. may be supplied by the function.

25. Write a declaration for a function called blyth() that takes two arguments and returns type char. The first argument is type int, and the second is type float with a default value of 3.14159.

Ans: `char blyth(int, float = 3.14159);`

26. Scope and storage class are concerned with the **<u>Visibility</u>** and **<u>Lifetime</u>** of a variable.

27. What functions can access a global variable that appears in the same file with them?

Ans: All functions in the same file

28. What functions can access a local variable?

Ans: Only the function (or block) in which it is declared

29. A static local variable is used to

a. make a variable visible to several functions.

b. make a variable visible to only one function.

c. conserve memory when a function is not executing.

d. retain a value when a function is not executing.

Ans: d. retain a value when a function in not executing.

30. In what unusual place can you use a function call when a function returns a value by reference?

Ans: One the left side of an assignment operator(=).

```
Ex:    getValue() = 100;
```

Where `getValue()` **returns a reference** `(int&)`